

```
"""
```

ledger_publisher.py – Append signed certification to the public ledger.

Specification: GRUS-SEAL-001-v1.0 Section 4

Append-only public ledger; mirrors to GitHub, Zenodo, Internet Archive.

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```
"""
```

```
import os, yaml
```

```
from typing import Dict
```

```
from dataclasses import dataclass
```

```
@dataclass
```

```
class LedgerPublishResult:
```

```
    passed: bool
```

```
    ledger_entry_path: str = ""
```

```
    ledger_url: str = ""
```

```
    failure_reason: str = ""
```

```
def publish_to_ledger(certification_record: Dict,
```

```
                      ledger_dir: str = "./grus-ledger") -> LedgerPublishResult:
```

```
    audit_id = certification_record.get("audit_id", "")
```

```
    if not audit_id:
```

```
        return LedgerPublishResult(passed=False, failure_reason="audit_id missing")
```

```
    audits_dir = os.path.join(ledger_dir, "audits")
```

```
    os.makedirs(audits_dir, exist_ok=True)
```

```
    out_path = os.path.join(audits_dir, f"{audit_id}.yaml")
```

```
    if os.path.exists(out_path):
```

```
        return LedgerPublishResult(passed=False,
```

```
                                   failure_reason=f"Audit {audit_id} already exists in ledger (append-only violation)")
```

```
    with open(out_path, "w", encoding="utf-8") as f:
```

```
        yaml.safe_dump(certification_record, f, sort_keys=False, default_flow_style=False)
```

```
    ledger_url = f"https://ledger.grus.utility/audits/{audit_id}"
```

```
    return LedgerPublishResult(passed=True, ledger_entry_path=out_path, ledger_url=ledger_url)
```

```
if __name__ == "__main__":
```

```
    import sys
```

```
    if len(sys.argv) < 2:
```

```
        print("Usage: python ledger_publisher.py <certification_record.yaml>"); sys.exit(0)
```

```
    with open(sys.argv[1]) as f:
```

```
        record = yaml.safe_load(f)
```

```
    r = publish_to_ledger(record)
```

```
    if r.passed:
```

```
        print(f"[PUBLISHED] {r.ledger_entry_path}")
```

```
        print(f"  URL: {r.ledger_url}")
```

```
    else:
```

```
        print(f"[FAIL] {r.failure_reason}"); sys.exit(1)
```